

Cubic regularization for box-constrained order-value optimization

A draft of the modified theory

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Abstract

This note develops the complexity theory for the variant of the regularized Newton-type method for order-value optimization in which the Tikhonov term $\frac{\sigma}{2}\|x - \bar{x}\|^2$ of the model is replaced by the cubic term $\frac{\sigma}{3}\|x - \bar{x}\|^3$. The optimality condition $C_{\delta,\epsilon}$ and the outer structure of the method are unchanged, so the resulting bounds are directly comparable with those of the quadratically regularized method. Three things change, and all three are visible in the numerical experiments. First, the iteration complexity improves from $O(\delta^{-2}\epsilon^{-2})$ to $O(\delta^{-3}\epsilon^{-3/2})$, so that the cubic bound is the smaller of the two precisely when $\epsilon < \delta^2$. Second, the improved order requires the curvature matrices to *dominate* the component Hessians, $B_i \succeq \nabla^2 f_i$, which makes the curvature an essential ingredient rather than an optional one: the choice $B_i = 0$ is no longer covered. Third, the improvement is an improvement in outer iterations only; the number of function evaluations carries an extra $\log_\gamma \bar{\sigma}$ factor that grows with the regularization threshold, which is larger in the cubic case.

1 What changes and what does not

We keep the notation of the main paper. Given $f_1, \dots, f_m$ and $p \in \{1, \dots, m\}$, the order-value function f returns the p th smallest value of $f_1(x), \dots, f_m(x)$, the problem is Minimize $_{x \in \Omega} f(x)$ over the box $\Omega = \{x \mid \ell \leq x \leq u\}$, and for $\delta > 0$

$$I_\delta(x) := \{i \mid f(x) - \delta \leq f_i(x) \leq f(x) + \delta\}$$

is the active index set. Given $M > 0$, a family $\mathcal{B} = \{B_i\}_{i=1}^m$ of symmetric matrices with $B_i \succeq 0$ and $\|B_i\| \leq M$ is called *admissible*.

The *cubically regularized model* at $\bar{x} \in \Omega$ is

$$\Psi_c(x; \bar{x}, \mathcal{B}, \delta, \sigma) = \max_{i \in I_\delta(\bar{x})} \left\{ \nabla f_i(\bar{x})^T (x - \bar{x}) + \frac{1}{2} (x - \bar{x})^T B_i (x - \bar{x}) \right\} + \frac{\sigma}{3} \|x - \bar{x}\|^3. \quad (1)$$

Since the regularization does not depend on i , writing it inside or outside the maximum is the same function; the implementation puts it inside each component, which is what the constraint-wise formulation of the subproblem requires.

Remark 1.1. Ψ_c is convex, being a maximum of convex quadratics plus a convex term. It is strictly convex, because $x \mapsto \|x - \bar{x}\|^3$ is: if $d_1 \neq d_2$ then $\|\frac{d_1 + d_2}{2}\| \leq \frac{1}{2}(\|d_1\| + \|d_2\|)$ and $t \mapsto t^3$

is strictly convex on $[0, \infty)$, so equality forces $\|d_1\| = \|d_2\|$ together with equality in the triangle inequality, hence $d_1 = d_2$. Therefore the subproblem

$$\underset{x \in \Omega}{\text{Minimize}} \Psi_c(x; x^k, \mathcal{B}_{k,j}, \delta, \sigma_{k,j}) \quad (2)$$

still has a unique global minimizer and every stationary point is global. Note that Ψ_c is not strongly convex — its curvature degenerates at $x = \bar{x}$ — so the uniqueness argument of the quadratic case, which invoked strong convexity, has to be replaced by the above.

Algorithm 1.1 (cubic variant). Let $\delta > 0$, $\sigma_{\min} > 0$, $M > 0$, $\alpha \in (0, 1)$, $\gamma > 1$, and $x^0 \in \Omega$ be given. Initialize $k \leftarrow 0$.

Step 1. Initialize $j \leftarrow 0$ and $\sigma_{k,j} = \sigma_{\min}$.

Step 2. Choose a curvature family $\mathcal{B}_{k,j}$ satisfying Assumption A6 below and compute $x_{\text{trial}}^{k,j}$ as the global minimizer of (2).

Step 3. If

$$f(x_{\text{trial}}^{k,j}) \leq f(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^3 \quad (3)$$

does not hold, set $\sigma_{k,j+1} = \gamma \sigma_{k,j}$, $j \leftarrow j + 1$, and go to Step 2.

Step 4. Set $x^{k+1} = x_{\text{trial}}^{k,j}$, $\sigma_k = \sigma_{k,j}$, $j_k = j$, $k \leftarrow k + 1$, and go to Step 1.

Throughout we write $d^{k,j} := x_{\text{trial}}^{k,j} - x^k$. Two changes with respect to the quadratic method deserve emphasis. The acceptance test (3) is *cubic* in the step: this is what matches the order of the model error under a Lipschitz Hessian, and it is what makes Theorem 3.2 below work. And Step 2 no longer accepts an arbitrary admissible family: the curvature must dominate the component Hessians, in the sense of Assumption A6.

Everything else is untouched. In particular the following results, which concern only the sign structure of the model, carry over verbatim from the main paper.

Theorem 1.2. Suppose that $f_1, \dots, f_m \in C^1(\Omega)$, let $\mathcal{B}$ be admissible and $\sigma, \delta \geq 0$. If x^* is a local minimizer of $\text{Minimize}_{x \in \Omega} f(x)$, then x^* is a local minimizer of $\text{Minimize}_{x \in \Omega} \Psi_c(x; x^*, \mathcal{B}, \delta, \sigma)$.

Proof. Identical to the chain of Theorems 3.1–3.3 of the main paper. The only place where the shape of the regularization enters is the step in which one argues that, if some y^* makes every component of the model negative, then already the linear parts are negative: there one uses that $\frac{1}{2}(y^* - x^*)^T B_i(y^* - x^*) \geq 0$ because $B_i \succeq 0$ and that the regularization term is nonnegative. Since $\frac{\sigma}{3}\|y^* - x^*\|^3 \geq 0$ for $\sigma \geq 0$, the argument is unchanged. $\square$

Consequently the exact optimality condition of Corollary 3.4 and the approximate condition $C_{\delta, \epsilon}$ of Definition 3.5 are *literally the same* for both methods: there exist $\mu \in \Sigma_{|I_\delta(x)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ with $(\ell_i - x_i)\nu_i^\ell = (x_i - u_i)\nu_i^u = 0$ and

$$\left\| \sum_{i \in I_\delta(x)} \mu_i \nabla f_i(x) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i \right\| \leq \epsilon. \quad (4)$$

This is what makes the two complexity bounds comparable: they count iterations until the same quantity falls below the same threshold.

2 The step equation and the step bound

Assumptions A1–A3 are those of the main paper.

Assumption A1. $f_1, \dots, f_m \in \mathcal{C}^1(\Omega)$.

Assumption A2. The multipliers of Corollary 2.1 satisfy $\max_i |\nu_i^\ell - \nu_i^u| \leq c_\nu$ uniformly in k and j .

Assumption A3. $\|\nabla f_i(x)\| \leq c_\nabla$ for all i and all $x \in \Omega$.

Corollary 2.1. Under Assumption A1, for every k and $j = 0, \dots, j_k$ there exist $\mu \in \Sigma_{|I_\delta(x^k)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ with $(\ell_i - [x_{\text{trial}}^{k,j}]_i)\nu_i^\ell = ([x_{\text{trial}}^{k,j}]_i - u_i)\nu_i^u = 0$ and

$$(\bar{B}_{k,j} + \sigma_{k,j}\|d^{k,j}\|I)d^{k,j} = \sum_{i=1}^n (\nu_i^\ell - \nu_i^u)e_i - \sum_{i \in I_\delta(x^k)} \mu_i \nabla f_i(x^k), \quad (5)$$

where $\bar{B}_{k,j} := \sum_{i \in I_\delta(x^k)} \mu_i B_{k,j,i}$ is symmetric positive semidefinite with $\|\bar{B}_{k,j}\| \leq M$.

Proof. As in Corollary 3.6 of the main paper, the KKT conditions for the minimization of a maximum of differentiable functions over a box give

$$\sum_{i \in I_\delta(x^k)} \mu_i [\nabla f_i(x^k) + B_{k,j,i}d^{k,j}] + \sigma_{k,j}\|d^{k,j}\|d^{k,j} + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell)e_i = 0.$$

The function $d \mapsto \frac{\sigma}{3}\|d\|^3$ is continuously differentiable on all of $\mathbb{R}^n$ with gradient $\sigma\|d\|d$, which vanishes at $d = 0$; no nonsmoothness is introduced at the reference point. Since $\sum_i \mu_i = 1$, the curvature terms collect into $\bar{B}_{k,j}d^{k,j}$, and $\bar{B}_{k,j}$, a convex combination of symmetric positive semidefinite matrices of norm at most M , has the stated properties. $\square$

The difference with the quadratic case is now visible: the matrix acting on the step is $\bar{B}_{k,j} + \sigma_{k,j}\|d^{k,j}\|I$ rather than $\bar{B}_{k,j} + \sigma_{k,j}I$. Its smallest eigenvalue degenerates with the step, and the resulting step bound is correspondingly weaker.

Lemma 2.2. Under Assumptions A1–A3,

$$\|d^{k,j}\| \leq \sqrt{\frac{c_x}{\sigma_{k,j}}}, \quad c_x := n c_\nu + c_\nabla. \quad (6)$$

Proof. The right-hand side of (5) has norm at most $\sum_{i=1}^n |\nu_i^\ell - \nu_i^u| + \sum_i \mu_i \|\nabla f_i(x^k)\| \leq n c_\nu + c_\nabla = c_x$. For the left-hand side, if $d^{k,j} \neq 0$ then by Cauchy–Schwarz and $\bar{B}_{k,j} \succeq 0$,

$$\begin{aligned} \|(\bar{B}_{k,j} + \sigma_{k,j}\|d^{k,j}\|I)d^{k,j}\| \|d^{k,j}\| &\geq \langle (\bar{B}_{k,j} + \sigma_{k,j}\|d^{k,j}\|I)d^{k,j}, d^{k,j} \rangle \\ &= (d^{k,j})^T \bar{B}_{k,j} d^{k,j} + \sigma_{k,j}\|d^{k,j}\|^3 \geq \sigma_{k,j}\|d^{k,j}\|^3, \end{aligned}$$

so that $\|(\bar{B}_{k,j} + \sigma_{k,j}\|d^{k,j}\|I)d^{k,j}\| \geq \sigma_{k,j}\|d^{k,j}\|^2$. Combining, $\sigma_{k,j}\|d^{k,j}\|^2 \leq c_x$, which is (6); the case $d^{k,j} = 0$ is trivial. $\square$

Remark 2.3. In the quadratic method the corresponding bound is $\|d^{k,j}\| \leq c_x/\sigma_{k,j}$. The cubic step therefore shrinks only like $\sigma_{k,j}^{-1/2}$. Since the mechanism that keeps the trial point inside the δ -band is precisely the smallness of the step, this is the reason why the regularization threshold below deteriorates from δ^{-1} to δ^{-2} .

3 Sufficient descent

The quadratic theory needed only a Lipschitz gradient. The cubic term is of higher order than the quadratic model error $\frac{L}{2}\|d\|^2$ and cannot absorb it for small steps, so a Lipschitz gradient is no longer enough: we need the model to be a genuine second-order model, with a Lipschitz Hessian and with curvature matrices that dominate the true ones.

Assumption A4. *There is $L > 0$ with $\|\nabla f_i(y) - \nabla f_i(x)\| \leq L\|y - x\|$ for all i and $x, y \in \Omega$.*

Assumption A5. *$f_1, \dots, f_m \in \mathcal{C}^2(\Omega)$ and there is $L_H > 0$ with $\|\nabla^2 f_i(y) - \nabla^2 f_i(x)\| \leq L_H\|y - x\|$ for all i and $x, y \in \Omega$.*

Assumption A6 (dominating curvature). *For all k , all $j = 0, \dots, j_k$ and all $i \in I_\delta(x^k)$,*

$$B_{k,j,i} \succeq \nabla^2 f_i(x^k), \quad B_{k,j,i} \succeq 0, \quad \|B_{k,j,i}\| \leq M.$$

Assumption A5 gives the standard cubic Taylor bound

$$f_i(y) \leq f_i(x) + \nabla f_i(x)^T(y - x) + \frac{1}{2}(y - x)^T \nabla^2 f_i(x)(y - x) + \frac{L_H}{6}\|y - x\|^3. \quad (7)$$

Remark 3.1 (on Assumption A6). *The shifted exact Hessian*

$$B_{k,j,i} = \nabla^2 f_i(x^k) + \tau_{k,i}I, \quad \tau_{k,i} := \max\{0, -\lambda_{\min}(\nabla^2 f_i(x^k))\},$$

satisfies Assumption A6 with $M := \max_i \sup_{x \in \Omega} (\|\nabla^2 f_i(x)\| + \max\{0, -\lambda_{\min}(\nabla^2 f_i(x))\})$, which is finite because Ω is compact and $f_i \in \mathcal{C}^2$. This is exactly the matrix built by the implementation.

Two consequences must be stated plainly. First, M is no longer a free parameter that the method may impose: it is a quantity determined by the problem. Rescaling $B_{k,j,i}$ by $M/\lambda_{\max}$ to force a prescribed bound, as the current implementation does, destroys the domination $B_{k,j,i} \succeq \nabla^2 f_i(x^k)$ whenever the rescaling is active, and with it Theorem 3.2. Second, and more importantly, the choice $B_{k,j,i} = 0$ is admissible only if $\nabla^2 f_i(x^k) \preceq 0$. The first-order method is therefore not a special case of the theory of this note: with cubic regularization, curvature is not an optional refinement but a requirement of the analysis. Quasi-Newton families such as BFGS or SR1 are admissible only insofar as they can be certified to dominate the component Hessians, which the plain updates do not guarantee.

Theorem 3.2 (sufficient descent). *Suppose that Assumptions A1, A3, A4, A5 and A6 hold.*

Then, for all k and $j = 0, \dots, j_k$, if

$$\sigma_{k,j} \geq \Theta(\delta) := \max\left\{\frac{L_H}{2} + 3\alpha, \frac{36c_x^3}{\delta^2}, \frac{3Lc_x}{\delta}\right\}, \quad (8)$$

then $x_{\text{trial}}^{k,j}$ satisfies (3).

Proof. Write $d := d^{k,j}$ and $\sigma := \sigma_{k,j}$.

Step 1: descent for the active components. Let $i \in I_\delta(x^k)$. By (7) and then Assumption A6,

$$\begin{aligned} f_i(x_{\text{trial}}^{k,j}) &\leq f_i(x^k) + \nabla f_i(x^k)^T d + \frac{1}{2}d^T \nabla^2 f_i(x^k)d + \frac{L_H}{6}\|d\|^3 \\ &\leq f_i(x^k) + \nabla f_i(x^k)^T d + \frac{1}{2}d^T B_{k,j,i}d + \frac{L_H}{6}\|d\|^3. \end{aligned}$$

132 Since $x_{\text{trial}}^{k,j}$ minimizes (2) and $\Psi_c(x^k; x^k, \mathcal{B}_{k,j}, \delta, \sigma) = 0$, we have $\Psi_c(x_{\text{trial}}^{k,j}; \cdot) \leq 0$; as Ψ_c bounds
 133 each of the components of its maximum,

$$134 \quad \nabla f_i(x^k)^T d + \frac{1}{2} d^T B_{k,j,i} d + \frac{\sigma}{3} \|d\|^3 \leq 0 \quad \text{for all } i \in I_\delta(x^k).$$

135 Substituting,

$$136 \quad f_i(x_{\text{trial}}^{k,j}) \leq f_i(x^k) - \left(\frac{\sigma}{3} - \frac{L_H}{6} \right) \|d\|^3 \leq f_i(x^k) - \alpha \|d\|^3, \quad i \in I_\delta(x^k), \quad (9)$$

137 the last inequality because (8) gives $\sigma \geq \frac{L_H}{2} + 3\alpha$, i.e. $\frac{\sigma}{3} - \frac{L_H}{6} \geq \alpha$.

138 *Step 2: the active set does not move out of the band.* By Assumption A4 and Cauchy–
 139 Schwarz, for every $i \in \{1, \dots, m\}$,

$$140 \quad |f_i(x_{\text{trial}}^{k,j}) - f_i(x^k)| \leq \|\nabla f_i(x^k)\| \|d\| + \frac{L}{2} \|d\|^2 \leq c_x \|d\| + \frac{L}{2} \|d\|^2,$$

141 using $c_\nabla \leq c_x$. By Lemma 2.2, $\|d\| \leq (c_x/\sigma)^{1/2}$, hence

$$142 \quad |f_i(x_{\text{trial}}^{k,j}) - f_i(x^k)| \leq \frac{c_x^{3/2}}{\sigma^{1/2}} + \frac{L c_x}{2\sigma}.$$

143 By (8), $\sigma \geq 36c_x^3/\delta^2$ gives $c_x^{3/2}\sigma^{-1/2} \leq \delta/6$, and $\sigma \geq 3Lc_x/\delta$ gives $Lc_x/(2\sigma) \leq \delta/6$. Therefore

$$144 \quad |f_i(x_{\text{trial}}^{k,j}) - f_i(x^k)| \leq \frac{\delta}{3}, \quad i = 1, \dots, m. \quad (10)$$

145 *Step 3: transfer to the order-value function.* From (10), any index with $f_i(x^k) < f(x^k) - \delta$
 146 satisfies $f_i(x_{\text{trial}}^{k,j}) < f(x^k)$ and any index with $f_i(x^k) > f(x^k) + \delta$ satisfies $f_i(x_{\text{trial}}^{k,j}) > f(x^k)$;
 147 in particular $i_p(x_{\text{trial}}^{k,j}) \in I_\delta(x^k)$ and the number of components below the band is the same at
 148 x^k and at $x_{\text{trial}}^{k,j}$. Writing $I_\delta(x^k) = \{i_1, \dots, i_r\}$ with $f_{i_1}(x^k) \leq \dots \leq f_{i_r}(x^k)$ and reordering as
 149 $\{i'_1, \dots, i'_r\}$ so that $f_{i'_1}(x_{\text{trial}}^{k,j}) \leq \dots \leq f_{i'_r}(x_{\text{trial}}^{k,j})$, the indices $i_p(x^k)$ and $i_p(x_{\text{trial}}^{k,j})$ occupy the same
 150 position q within the band. Applying Lemma 2.1 of [1] with $a_j = f_{i_j}(x^k)$, $b_j = f_{i_j}(x_{\text{trial}}^{k,j})$ and
 151 $\beta = \alpha \|d\|^3$ —legitimate by (9)—yields $f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f_{i_q}(x^k) - \beta$, that is, $f(x_{\text{trial}}^{k,j}) \leq f(x^k) - \alpha \|d\|^3$,
 152 which is (3). $\square$

153 Note that Step 3 is where the cubic form of the acceptance test (3) is forced: the ordering
 154 lemma transfers exactly the constant β obtained in (9) from the components to f , and that
 155 constant is $\alpha \|d\|^3$.

156 4 Complexity

157 **Theorem 4.1.** Suppose that Assumptions A1, A3–A6 hold and that $f(x) \geq f_{\text{low}}$ on Ω . Put
 158 $\bar{\sigma} := \gamma \Theta(\delta)$ with $\Theta(\delta)$ as in (8). Then $\sigma_k \leq \bar{\sigma}$ for every k , at most

$$159 \quad \left\lceil 1 + \log_\gamma(\sigma_k/\sigma_{\min}) \right\rceil \quad (11)$$

160 function evaluations are performed per outer iteration, and the number of outer iterations at
 161 which $C_{\delta,\epsilon}$ fails at x^{k+1} is at most

$$162 \quad \left\lfloor \frac{f(x^0) - f_{\text{low}}}{\alpha} \max \left\{ \left(\frac{2M}{\epsilon} \right)^3, \left(\frac{2\bar{\sigma}}{\epsilon} \right)^{3/2} \right\} \right\rfloor. \quad (12)$$

163 *Proof.* The bound $\sigma_k \leq \bar{\sigma}$ and (11) follow from Theorem 3.2 and the update $\sigma_{k,j+1} = \gamma\sigma_{k,j}$
 164 exactly as in the quadratic case: starting from $\sigma_{k,0} = \sigma_{\min}$, at most $\lfloor 1 + \log_{\gamma}(\sigma_k/\sigma_{\min}) \rfloor$ increases
 165 occur before (8) holds, and one function evaluation is spent per increase.

166 Let K be the set of outer indices at which $C_{\delta,\epsilon}$ fails at x^{k+1} , fix $k \in K$, and write $d_k :=$
 167 $x^{k+1} - x^k$, $\bar{B}_k := \bar{B}_{k,j_k}$. By Corollary 2.1 and the failure of (4),

$$168 \quad \epsilon < \left\| \sum_{i \in I_{\delta}(x^k)} \mu_i \nabla f_i(x^k) + \sum_{i=1}^n (\nu_i^u - \nu_i^{\ell}) e_i \right\| = \|(\bar{B}_k + \sigma_k \|d_k\| I) d_k\| \leq M \|d_k\| + \sigma_k \|d_k\|^2.$$

169 Hence at least one of $M \|d_k\| > \epsilon/2$ and $\sigma_k \|d_k\|^2 > \epsilon/2$ holds, so that, using $\sigma_k \leq \bar{\sigma}$,

$$170 \quad \|d_k\| > \min \left\{ \frac{\epsilon}{2M}, \sqrt{\frac{\epsilon}{2\bar{\sigma}}} \right\} =: s_{\epsilon}.$$

171 Since x^{k+1} satisfies (3), $f(x^{k+1}) \leq f(x^k) - \alpha \|d_k\|^3 < f(x^k) - \alpha s_{\epsilon}^3$. Summing over $k \in K$ and
 172 using $f \geq f_{\text{low}}$ gives $f(x^0) - f_{\text{low}} > |K| \alpha s_{\epsilon}^3$, and $s_{\epsilon}^{-3} = \max\{(2M/\epsilon)^3, (2\bar{\sigma}/\epsilon)^{3/2}\}$ yields (12). $\square$

173 The bound (12) has two branches, and which one is active is decided by the size of the
 174 curvature relative to the target accuracy.

175 **Corollary 4.2.** *Under the hypotheses of Theorem 4.1, if*

$$176 \quad \epsilon \geq \frac{2M^2}{\bar{\sigma}}, \quad (13)$$

177 *then the number of outer iterations at which $C_{\delta,\epsilon}$ fails at x^{k+1} is at most*

$$178 \quad \left\lfloor \frac{f(x^0) - f_{\text{low}}}{\alpha} \frac{(2\bar{\sigma})^{3/2}}{\epsilon^{3/2}} \right\rfloor = O\left(\delta^{-3} \epsilon^{-3/2}\right). \quad (14)$$

179 *Proof.* $(2M/\epsilon)^3 \leq (2\bar{\sigma}/\epsilon)^{3/2}$ is equivalent, on raising both sides to the power 2/3, to $(2M)^2 \leq$
 180 $2\bar{\sigma}\epsilon$, i.e. to (13); the first branch of (12) is then inactive. The order statement follows from
 181 $\bar{\sigma} = \gamma\Theta(\delta)$ and $\Theta(\delta) = O(\delta^{-2})$, so $\bar{\sigma}^{3/2} = O(\delta^{-3})$. $\square$

182 Comparison with the quadratic method

183 Table 1 collects the two analyses. The optimality condition, the outer loop and the inner σ -loop
 184 are identical, so the entries are directly comparable.

185 Three consequences are worth isolating.

	quadratic $\frac{\sigma}{2}\ d\ ^2$	cubic $\frac{\sigma}{3}\ d\ ^3$
acceptance test	$f(x^k) - \alpha\ d\ ^2$	$f(x^k) - \alpha\ d\ ^3$
smoothness needed	∇f_i Lipschitz (L)	$\nabla^2 f_i$ Lipschitz (L_H), and ∇f_i Lipschitz
curvature needed	$B_i \succeq 0, \ B_i\ \leq M$	$B_i \succeq \nabla^2 f_i, B_i \succeq 0, \ B_i\ \leq M$
$B_i = 0$ admissible	yes	only if $\nabla^2 f_i \preceq 0$
step bound	$\ d\ \leq c_x/\sigma$	$\ d\ \leq (c_x/\sigma)^{1/2}$
σ threshold $\Theta(\delta)$	$O(\delta^{-1})$	$O(\delta^{-2})$
outer iterations	$O(\delta^{-2}\epsilon^{-2})$	$O(\delta^{-3}\epsilon^{-3/2})$ if $\epsilon \geq 2M^2/\bar{\sigma}$
evaluations / iteration	$\lfloor 1 + \log_\gamma(\sigma_k/\sigma_{\min}) \rfloor$	$\lfloor 1 + \log_\gamma(\sigma_k/\sigma_{\min}) \rfloor$, larger σ_k

Table 1: The quadratically and cubically regularized methods, measured against the same approximate optimality condition $C_{\delta,\epsilon}$.

186 **(i) The cubic bound is smaller exactly when $\epsilon < \delta^2$.** Ignoring constants, the two iteration
187 bounds are $\delta^{-2}\epsilon^{-2}$ and $\delta^{-3}\epsilon^{-3/2}$, whose ratio is

$$188 \quad \frac{\delta^{-3}\epsilon^{-3/2}}{\delta^{-2}\epsilon^{-2}} = \frac{\epsilon^{1/2}}{\delta},$$

189 which is less than one if and only if $\epsilon < \delta^2$. The cubic regularization buys accuracy in ϵ and pays
190 for it in δ : the narrower the active band, the larger the regularization needed to keep the trial
191 point inside it, and that cost is amplified by the exponent 3/2. The parameters of the numerical
192 section, $\delta = 10^{-1}$ and $\epsilon = 10^{-4}$, satisfy $\epsilon < \delta^2 = 10^{-2}$ comfortably, which is the regime in which
193 the cubic bound is the better of the two.

194 **(ii) The curvature is what limits the improved regime.** In the quadratic analysis the
195 bound M on the curvature enters only through the constant $(M + \gamma\Theta(\delta))^2$, because M and σ
196 multiply the step in the same way. Under cubic regularization the residual is $M\|d\| + \sigma\|d\|^2$,
197 so the two terms scale differently and M enters the *order*: below the threshold $\epsilon < 2M^2/\bar{\sigma}$ the
198 bound degrades to $O(\epsilon^{-3})$. The improved rate is therefore a moderate-accuracy statement, valid
199 on the window

$$200 \quad \frac{2M^2}{\bar{\sigma}} \leq \epsilon < \delta^2,$$

201 which is nonempty whenever $2M^2 < \bar{\sigma}\delta^2$; with $\bar{\sigma} \geq 36\gamma c_x^3\delta^{-2}$ this holds as soon as $M^2 < 18\gamma c_x^3$,
202 a mild condition. One would like to shrink M to widen the window, but Assumption A6 forbids
203 it: M must be large enough for the B_i to dominate the component Hessians. This is a genuine
204 tension of the cubic variant and it should be stated rather than hidden.

205 **(iii) The gain is in iterations, not in evaluations.** The total number of function evaluations
206 is at most the number of outer iterations times $\lfloor 1 + \log_\gamma(\bar{\sigma}/\sigma_{\min}) \rfloor$. The cubic method has a
207 larger $\bar{\sigma} = O(\delta^{-2})$ against $O(\delta^{-1})$ — so it spends more inner increases of σ per outer iteration,
208 and the improvement in evaluations is smaller than the improvement in iterations by a factor of
209 order $\log_\gamma(\Theta_{\text{cub}}/\Theta_{\text{quad}}) = \log_\gamma O(\delta^{-1})$. This is precisely the pattern of the experiments: on the
210 nonlinear problem the proposed method improves the mean iteration count by 4.44× but the
211 mean evaluation count only by 2.76×, and on the problem affine in the parameters it improves
212 iterations by 1.22× while *losing* 8% in evaluations.

5 What the implementation would have to change

Three points connect the theory above with the code that produced the numbers. The first has been settled; the remaining two must be reconciled before the numbers can be presented as illustrating these theorems.

1. **The acceptance test** (*done*). Step 3 of the implementation now uses the cubic test (3) in the second-order branch, and retains the quadratic test $f(x_{\text{trial}}^{k,j}) \leq f(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2$ in the first-order branch, which is the published method. Both problems were rerun after the change: the mean iteration and evaluation counts are *identical* to those obtained with the quadratic test, on every value of α and on both problems. This is what one expects, since $\alpha = 10^{-8}$ and $\|d\| < 1$ throughout, so the two right-hand sides differ by less than the gap between consecutive accepted and rejected trial points; but it is worth having verified rather than argued.
2. **The rescaling of the curvature.** The implementation shifts the exact Hessian to positive definiteness and then rescales it by $M/\lambda_{\max}$ when $\lambda_{\max} > M$, with $M = 1$. The shift is what Assumption A6 asks for; the rescaling breaks it. Either the rescaling is removed and M is redefined as in Remark 3.1 as the bound that the shifted Hessians happen to satisfy on Ω , or Theorem 3.2 does not apply to the runs. Removing it changes the numbers and the experiments would have to be rerun.
3. **The first-order comparison.** The baseline in the experiments is the quadratically regularized method with $B_i = 0$, which is covered by the theory of the main paper but not by the theory of this note. That is the correct comparison to report—it is the published method—but the two columns of the tables are then governed by two different sets of assumptions, and the text must say so. In particular, the cubically regularized method with $B_i = 0$, which performed best of all in the exploratory runs, is not covered by either analysis.

References

- [1] Order-value optimization ordering lemma; see Lemma 2.1 of the reference cited as **dunder1** in the main paper.
- [2] The first-order regularized method for order-value optimization generalized by the main paper.